Fiscal Year:	FY 2025	**Task Last Updated:**	FY 11/24/2025
PI Name:	Macias, Brandon Ph.D.		
Project Title:	Mechanical and Gravitational Countermeasures to Ocular Changes During Strict Head-Down Tilt Bedrest		
Division Name:	Human Research		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **HHC**:Human Health Countermeasures		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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PI Organization Type:	NASA CENTER	**Phone:**	281-483-2026
Organization Name:	NASA Johnson Space Center		
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City:	Houston	**State:**	TX
Zip Code:	77058	**Congressional District:**	36
Comments:	NOTE: Became civil servant fall 2020; previously KBR/NASA Johnson Space Center. Prior to that until 2016, was at the University of California, San Diego.		
Project Type:	Ground	**Solicitation / Funding Source:**	Directed Research
Start Date:	08/26/2022	**End Date:**	09/30/2025
No. of Post Docs:	1	**No. of PhD Degrees:**	
No. of PhD Candidates:		**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA JSC
Contact Monitor:	Brocato, Becky	**Contact Phone:**	
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Flight Program:			
Flight Assignment:	NOTE: End date changed to 09/30/2025 per C. Ribeiro/NASA-JSC. (Ed., 11/27/23). Original end date was 08/20/2024. NOTE: Project start date updated due to P.I. change to B. Macias. Information per C. Ribeiro/NASA-JSC. (Ed. 8/29/22).		
Key Personnel Changes/Previous PI:	na		
COI Name (Institution):	Brunstetter, Tyson O.D., Ph.D. (NASA Johnson Space Center) Huang, Alex M.D., Ph.D. (Doheny Eye Institute) Kramer, Larry M.D. (University of Texas Health Science Center at Houston) Laurie, Steven Ph.D. (KBR/NASA Johnson Space Center) Lee, Stuart Ph.D. (KBR/NASA Johnson Space Center) Lovering, Andrew Ph.D. (University of Oregon) Martin, Bryn Ph.D. (Alcyone Therapeutics Inc.) Young, Millennia Ph.D. (NASA Johnson Space Center) Lytle, Jason (KBR/NASA Johnson Space Center)		
Grant/Contract No.:	Directed Research		
Performance Goal No.:			

Performance Goal Text:	
Task Description:	The Spaceflight Associated Neuro-ocular Syndrome (SANS) is associated with structural and functional ocular and brain changes and develops in ~2/3rds of astronauts during long-duration spaceflight. Although the etiology of SANS is unknown, a weightlessness-driven headward fluid redistribution relative to the upright position on Earth is hypothesized to be a primary initiating factor. A countermeasure that can successfully reverse this headward fluid shift for multiple hours per day may be needed to prevent SANS. Exposure to 30 days of the spaceflight analog strict 6 degree head-down tilt (HDT) bedrest induces the development of optic disc edema, a hallmark sign of SANS, in 2/3 of subjects; thus, NASA will utilize this platform to test the effectiveness of various countermeasures to prevent or reduce the development of SANS findings. This study will include a control group exposed to 30 days of strict 6 degree HDT. A second group of subjects will be exposed to 6 hours of 25 mmHg lower body negative pressure (LBNP), daily, during 30 days of strict 6 degree HDT bedrest. A third group of subjects will be exposed to 6 hours of upright posture per day, and 6 degree HDT for the remaining 18 h/day. This will help determine if daily exposure to 6 hours total of upright posture, fully reversing the headward fluid for the same duration of use as the LBNP group, is sufficient to prevent SANS findings, and thus can direct future countermeasure studies with respect to the duration needed to successfully prevent SANS. We will assess multiple cerebral, vascular, and ocular outcome measures to determine the effects of 30 days strict 6 degree HDT bedrest, and the ability of daily 6 hour exposure to LBNP or upright positioning to prevent these changes. Specific Aims: 1. To determine if daily exposure to actual (upright posture) and simulated (LBNP) hydrostatic pressure gradients prevents ocular and cerebral structural changes during 30 days of strict 6° head-down tilt bedrest. 2. To determine if daily exposure to actual (upright posture) and simulated (LBNP) hydrostatic pressure gradients prevents ocular functional changes during 30 days of strict 6° head-down tilt bedrest. 3. To determine if daily exposure to actual (upright posture) and simulated (LBNP) hydrostatic pressure gradients prevents vascular and cerebrospinal fluid changes during 30 days of strict 6° head-down tilt bedrest. This bedrest study will be implemented in the :envihab facility at the German Aerospace Center (DLR) in Cologne, Germany.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Spaceflight associated neuro-ocular syndrome (SANS) develops in approximately 2/3 of astronauts during long-duration spaceflight. Although the etiology of SANS is unknown, a weightlessness-driven headward fluid redistribution relative to the upright position on Earth is hypothesized to be a primary initiating factor. A countermeasure that can successfully reverse this headward fluid shift for multiple hours per day may be needed to prevent SANS findings from developing. Exposure to 30 days of strict 6-degree head-down tilt (HDT) bedrest induces the development of optic disc edema, a hallmark sign of SANS, in approximately 2/3 of subjects; thus, NASA is utilizing this platform to test the effectiveness of various countermeasures to prevent or reduce the development of SANS findings.
Task Progress:	All data collection activities were completed across Campaigns 1 (September-November 2021), 2 (January-March 2022), 3 (January-March 2023), and 4 (May-July 2023). Personnel from the Cardiovascular and Vision Laboratory (CVL) traveled to the German Aerospace Center (DLR) five times to provide training and support before and during data collection. The purpose of the first visit was to provide in-person training sessions to DLR staff and to oversee baseline data collection (BDC) on the 12 subjects participating in the first campaign of the study. Additionally, due to unforeseen issues with the Dynamic Vessel Analyzer (DVA) hardware that resulted in the company being unable to train DLR staff prior to BDC as planned, a CVL team member collected all BDC data on the 12 subjects. CVL personnel then returned on an additional trip to collect recovery day 2 data (R+2) and continue training DLR staff to perform DVA measurements so that they could successfully collect data on R+12. CVL personnel then traveled back to DLR prior to subject arrival for the start of Campaign 3 to provide in-person training to DLR staff for the new protocol involving VTCs. During the visit, the subjects' BDC began, and the CVL was present to observe and guide the VTC fit checks to ensure that subjects were fit with the correct size of cuff. Another visit occurred mid-way through Campaign 3 to observe how subjects and operators were doing with the VTCs. The final visit was during the BDC timeframe of Campaign 4 to train new operators and observe the VTC fit checks. All data from the 4 bedrest campaigns have been imported into a study database and delivered to NASA Life Sciences Data Archive. The CVL also provided comprehensive lessons learned for future collaboration and bedrest studies at the facility. Six publications to date have resulted from this study. Their references and summaries are listed below. 1. Moestl S et al. (2025). Mechanical countermeasures for spaceflight-associated neuro-ocular syndrome during 30-days of head down tilt bed rest: design, implementation, and tolerability. Front Physiol; DOI: 10.3389/fphys.2025.1530783. 2. He Y, Karanjia R, Zhang X, Wanderer D, Walker E, Lee SH, Poczatek MJ, O'Grady CS, Marion KM, Young M, Moestl S, Mulder E, Stern C, Sadda SR, Laurie SS, Macias BR, Huang AS. Optic nerve vasculature and countermeasure assessment in a bedrest analogue of spaceflight-associated neuro-ocular syndrome. Am J Ophthalmol. 2025;278, 317–327. 3. Hoenemann J-N, Moestl S, de Boni L, Hoffmann F, Arz M, Berger L, Pesta D, Heusser K, Mulder E, Lee SMC, Macias BR, Tank J, Jordan J. Cardiopulmonary deconditioning and plasma volume loss are not sufficient to provoke orthostatic hypertension. Hypertens Res. 2024;1–6. 4. Hönemann J-N, Hoffmann F, de Boni L, Gauger P, Mulder E, Möstl S, Heusser K, Schmitz M-T, Halbach M, Laurie SS, Lee SMC, Macias BR, Jordan J. Tank J. Impact of daily lower-body negative pressure or cycling followed by venous constrictive thigh cuffs on bedrest-induced orthostatic intolerance. J Am Heart Assoc. 2024;13:e034800. 5. Berger L, Mulder E, Zange J, Frings-Meuthen P, Frett T, Simon P, Hoenemann J-N, Poczatek MJ, Laurie SS, Huang

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Bibliography Type:	Description: (Last Updated: 06/03/2026)
Articles in Peer-reviewed Journals	He Y, Karanjia R, Zhang X, Wanderer D, Walker E, Lee SH, Poczatek MJ, O'Grady CS, Marion KM, Young M, Moestl S, Mulder E, Stern C, Sadda SR, Laurie SS, Macias BR, Huang AS. "Optic Nnerve vasculature and countermeasure assessment in a bedrest analogue of spaceflight-associated neuro-ocular syndrome (SANS)." Am J Ophthalmol. 2025 Jun 20. http://dx.doi.org/10.1016/j.ajo.2025.06.032 ; PubMed PMID: 40545016; PubMed Central PMCID: PMC12462775 , Oct-2025
Articles in Peer-reviewed Journals	Hoenemann JN, Moestl S, de Boni L, Hoffmann F, Arz M, Berger L, Pesta D, Heusser K, Mulder E, Lee SMC, Macias BR, Tank J, Jordan J. "Cardiopulmonary deconditioning and plasma volume loss are not sufficient to provoke orthostatic hypertension." Hypertens Res. 2024 May 23. https://doi.org/10.1038/s41440-024-01710-x ; PubMed PMID: 38783144; PubMed Central PMCID: PMC11298404 , May-2024
Articles in Peer-reviewed Journals	Hönemann JN, Hoffmann F, de Boni L, Gauger P, Mulder E, Möstl S, Heusser K, Schmitz MT, Halbach M, Laurie SS, Lee SMC, Macias BR, Jordan J, Tank J. "Impact of daily lower-body negative pressure or cycling followed by venous constrictive thigh cuffs on bedrest-induced orthostatic intolerance." J Am Heart Assoc. 2024 Oct 25;e034800. https://doi.org/10.1161/JAHA.124.03480 ; PubMed PMID: 39450723; PubMed Central PMCID: PMC11935683 , Oct-2024
Articles in Peer-reviewed Journals	Berger L, Mulder E, Zange J, Frings-Meuthen P, Frett T, Simon P, Hoenemann JN, Poczatek MJ, Laurie SS, Huang AS, Macias B, Jordan J, Tank J, Rittweger J, Schmitz MT, Pesta D. "Aerobic exercise and subsequent venoconstrictive thigh cuffs preserve cardiorespiratory fitness but not muscle function after 30 days of bedrest." Med Sci Sports Exerc. 2025 Dec;57(12):2887-901. http://dx.doi.org/10.1249/MSS.0000000000003803 ; PubMed PMID: 40605173 , Dec-2025
Articles in Peer-reviewed Journals	Laurie SS, Lytle JR, Pickering SK, Poczatek MJ, Lee SMC, Pardon LP, Miller AE, Martin DS, Mulder E, Moestl S, Pesta D, Frett T, Stern C, Tank J, Hoenemann JN, Sadda S, Karanjia R, Marshall-Goebel K, Young M, Cole CW, Huang AS, Macias BR. "Daily use of countermeasures to prevent headward fluid shifts throughout 30 days of strict head-down tilt bedrest." J Appl Physiol (1985). 2025 Aug 1;139(2):394-413. https://doi.org/10.1152/japplphysiol.00635.2024 ; PubMed PMID: 40590841 , Aug-2025
Articles in Peer-reviewed Journals	Carter KJ, Laurie SS, Warthen KG, Sater SH, Martin BA, Feng CM, Young M, Hasan KM, Kramer LA, Macias BR. "Normal variation in brain volumetrics, CSF dynamics, and ocular structures from magnetic resonance images of healthy participants over two years." J Appl Physiol (1985). 2025 Jul 1;139(1):1-13. https://doi.org/10.1152/japplphysiol.00596.202 ; PubMed PMID: 40397516 , Jul-2025